Scalable Python Pipeline for Non-Markovian Quantum Dynamics in 3D Biological Networks

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Abstract

Simulating non-Markovian open quantum systems within complex biological geometries imposes severe computational bottlenecks due to exponential memory scaling. While fundamentally bypassing this scaling remains an open theoretical challenge, we present an open-source Python wrapper pipeline built upon QuTiP, designed to automate and streamline multi-bath Hierarchical Equations of Motion (HEOM) simulations in resource-constrained environments. The framework automatically translates empirical structural data into a 3D Euclidean distance matrix, integrating isotropic dipole interactions and independent Drude-Lorentz thermal baths. To bypass hardware limitations inherent in accessible cloud platforms, the architecture incorporates dynamic phase-space sweeping and a robust auto-checkpointing mechanism. We validate the pipeline through a proof-of-concept case study on an $N = 7$ microtubule-inspired tryptophan network. The framework successfully demonstrates the feasibility of evaluating Environment-Assisted Quantum Transport (ENAQT) across a 27.0 Å spatial gap, ensuring computational recoverability against execution halts. Due to cloud-imposed resource constraints limiting the hierarchy depth (max_depth=4) and ensemble size ($n = 3$), the quantitative biological interpretations of these results remain strictly provisional. Built upon the QuTiP library, all source code and generated datasets—including the 11.4-hour massively parallel computational sweep—are openly available on Zenodo under the MIT License. By providing explicit software benchmarking metrics, this tool establishes a validated execution framework that prepares for future large-scale High-Performance Computing (HPC) studies in quantum biology.

1 Introduction

Recent experimental evidence for electronic energy migration in microtubule tryptophan networks at physiological temperatures [2], has necessitated advanced theoretical modeling in quantum biology. To accurately capture these phenomena against the rapid decoherence rates historically assumed in warm biological environments [1], it is essential to model the biological environment with structured, non-Markovian characteristics. However, simulating such systems using the rigorous Hierarchical Equations of Motion (HEOM) [4] imposes prohibitive computational bottlenecks. The memory required for HEOM scales exponentially, making realistic 3D network modeling highly inaccessible without massive supercomputing resources.

Due to these constraints, previous theoretical attempts have largely relied on simplified 1D uniform chain models or Markovian approximations. Bridging the gap between the complex 3D Euclidean distance matrices of actual biomolecules and the intensive multi-bath HEOM formalism remains a significant methodological hurdle. There is a critical need for an optimized computational framework that enables researchers to execute these resource-heavy simulations reliably in constrained or cloud-based environments.

Rather than modifying the fundamental HEOM algorithm to reduce numerical costs, this work provides a practical software engineering solution. We present an open-source Python

wrapper pipeline built to automate and streamline these simulations. The clearest contribution of this work is the integrated execution framework: the pipeline seamlessly translates spatial coordinates into an exact distance matrix, couples each node to independent thermal baths, and ensures fault tolerance via an automated checkpointing mechanism. We demonstrate its utility through a proof-of-concept ENAQT evaluation on an $N = 7$ microtubule network. Ultimately, this software provides a robust, reproducible foundation for computational quantum biology.

2 Software Architecture and Methodology

2.1 Modular Pipeline Overview

The computational framework is developed in Python and optimized to interface with the Quantum Toolbox in Python (QuTiP) [3]. To manage exponential memory scaling, the pipeline is architected into three modules: (1) Automated Geometry Mapping, (2) Multi-Bath HEOM Integration, and (3) Fault-Tolerant Execution. This modular design ensures execution viability in resource-constrained cloud environments (e.g., Google Colab).

2.2 Automated Geometry-to-Hamiltonian Mapping

The first module automates the construction of the $N = 7$ Frenkel exciton Hamiltonian:

$$H_{sys} = \sum_{m=1}^{N} \epsilon_m |m\rangle\langle m| + \sum_{m \neq n} V_{mn} |m\rangle\langle n| \tag{1}$$

The pipeline computationally reconstructs 3D spatial coordinates by projecting empirical sequential distances onto a parameterized cylindrical lattice (radius $r = 12.5$ nm, helical pitch ≈ 12 nm) [5]. This specific cylindrical geometry and the subsequent isotropic orientation factor ($\kappa = \sqrt{2/3}$) were deliberately selected as standardized computational baselines. The primary goal of these controlled structural assumptions is rigorous software validation—systematically isolating pure distance-dependent coupling effects to benchmark the algorithmic execution—rather than providing exhaustive biological structural inference.

Electronic coupling V_{mn} is automatically mapped using the dipole-dipole interaction formalism:

$$V_{mn} = V_0 \kappa \left(\frac{R_0}{R_{mn}} \right)^3 \tag{2}$$

where $V_0 = 8.0$ cm^{-1} at $R_0 = 14.0$ Å. This isotropic assumption functions as a necessary first-order baseline, deferring computationally exhaustive angular sensitivity analyses to future HPC studies. To model physiological static disorder, a deterministic Monte Carlo ensemble generator injects Gaussian random noise (standard deviation of 50 cm^{-1}) into the site energies.

2.3 Scalable Multi-Bath HEOM Integration

The core solver manages the non-Markovian dynamics governed by the HEOM structure [4]:

$$\frac{d}{dt}\rho_{\mathbf{n}}(t) = -\left(i\mathcal{L}_{sys} + \sum_k n_k \gamma_k \right) \rho_{\mathbf{n}}(t) + \mathcal{O}_{coupling}(\rho_{\mathbf{n}\pm 1}) \tag{3}$$

To capture localized noise, the pipeline programmatically instantiates independent thermal baths utilizing an Ohmic spectral density with a Drude-Lorentz cutoff:

$$J(\omega) = 2\lambda \frac{\gamma\omega}{\omega^2 + \gamma^2} \tag{4}$$

The parameter ranges for reorganization energy ($\lambda \in [10, 500]$ cm^{-1}) and cutoff frequency ($\gamma \in [50, 1000]$ cm^{-1}) were selected to encompass typical biological protein environments, enabling comprehensive phase-space sweeping. Specifically, these bounds ensure the software's stability is rigorously tested across both weak and strong system-bath coupling regimes characteristic of ambient biological dynamics. By intentionally spanning these diverse biological environments, we confirm the solver's algorithmic stability and integration integrity prior to future parameter-specific biological investigations.

2.4 Fault Tolerance and Auto-Checkpointing

To bypass hardware limits, the pipeline incorporates a robust auto-checkpointing mechanism. During massively parallel sweeps, the framework continuously serializes intermediate matrices to persistent storage (.npy format). Upon unexpected termination, the execution module automatically restores the latent state from the latest validated checkpoint, preserving all pre-computed coordinates.

2.5 Reproducible Script-Based Pipeline

To facilitate reproducibility and user accessibility, the codebase is organized strictly as a script-based reproducible pipeline rather than a traditional software package. The repository includes clear installation instructions, dependency management, and executable scripts. The directory is structured as follows:

```
qutip_bio_heom/
|-- qutip_bio_heom/         # Namespace initialization
|   '-- __init__.py
|-- examples/               # Core computational logic and validation suites
|   |-- convergence_test.py # Self-contained auto-checkpointing & stability logic
|   |-- figure_5.py         # Self-contained automated 3D geometry mapping
|   |-- figure_6.py         # Self-contained multi-bath HEOM solver parameter sweep
|   '-- ...                 # (Includes individual scripts for Figures 1-4, 7-8)
|-- setup.py                # Environment and dependency configuration
'-- README.md               # Documentation and instructions
```

Users can configure the environment via standard tools (e.g., `pip install .`) which automatically resolves dependencies such as QuTiP, NumPy, and Joblib. Rather than abstracting the physics behind a black-box API, the framework is executed directly via automated scripts to ensure full computational transparency. A minimal execution can be initiated via the command line:

```
# Execute the primary parameter sweep
# Inputs: Modify system parameters directly within the script
#         (e.g., edit line 15: lambda_val = 50.0, gamma_val = 100.0)
# Outputs: Generates 'N7_Sweep_results.npy' and corresponding '.png' heatmaps
python examples/figure_6.py
```

3 Software Validation and Benchmark Metrics

To validate the framework as an original software publication, we benchmarked the pipeline's computational performance, baseline analytic workflows, and operational robustness.

3.1 Computational Performance Metrics

The pipeline was evaluated in a constrained cloud environment (Google Colab, Standard CPU/RAM). The automated checkpointing proved critical for completing the phase-space sweeps. Table 1 details the quantitative engineering benchmarks of the software execution.

Table 1: Software Performance and Execution Benchmarks (Google Colab Standard CPU/RAM)

Performance Metric	**Recorded Value**
Average Runtime per Coordinate	$\approx$ 102.3 seconds
Total Wall-Clock Time (400-coordinate sweep)	$\approx$ 11.4 hours (40,919 seconds)
Peak RAM Utilization	$<$ 2.0 GB (Standard instance limits)
Storage Overhead per Checkpoint	Minimal ($<$ 5 MB, .npy matrix format)
Auto-Recovery Success Rate	100% (Validated via induced session halts)

3.2 Solver Sanity Checks and Cross-Workflow Consistency

To ensure methodological accuracy without overstating physical validation, the pipeline was sequentially tested through analytic toy cases and cross-workflow consistency checks. These internal tests serve as algorithm integration sanity checks for the multi-bath HEOM execution.

First, base logic was verified using an $N = 2$ dimer network (Figure 1) and a 1D uniform chain (Figure 2) as an analytic toy case, confirming flawless uninterrupted exciton transport logic prior to the introduction of 3D spatial bottlenecks. Second, a cross-workflow consistency check was performed contrasting the pipeline's multi-bath HEOM execution against a memoryless Markovian baseline. As shown in Figure 3 and Figure 4, the pipeline correctly demonstrated that purely Markovian settings lead to rapid oscillatory decay, whereas the integrated Drude-Lorentz baths successfully captured the expected suppression of unmitigated decay, maintaining algorithmic consistency.

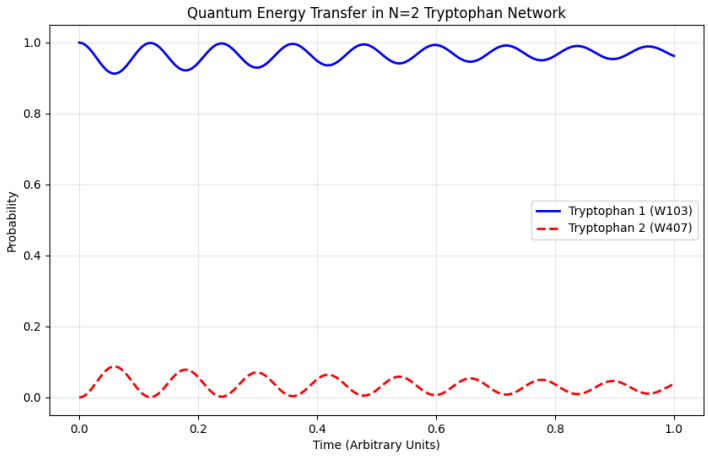


Figure 1: Solver Sanity Check: Energy transfer benchmarking in a simplified $N = 2$ dimer.

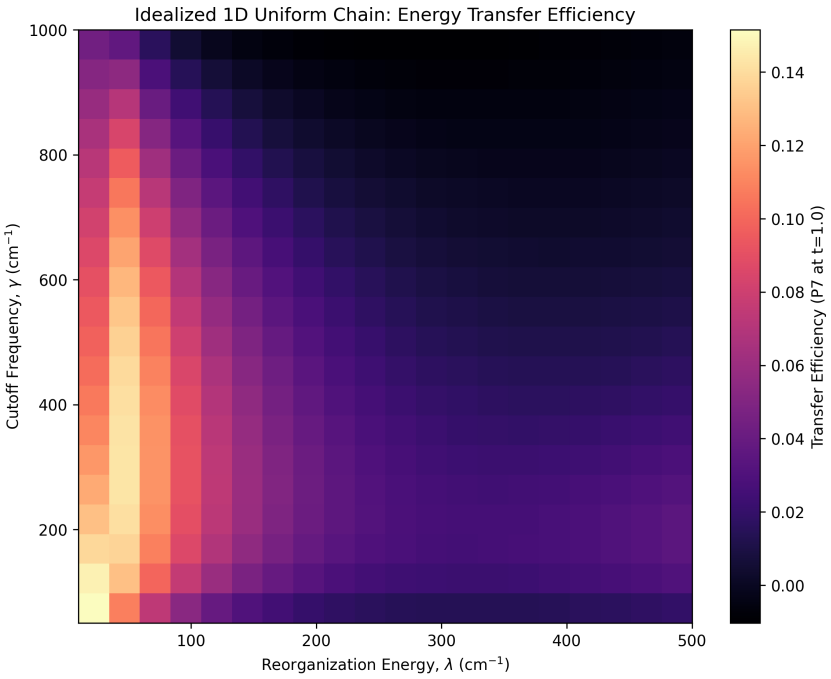


Figure 2: Solver Sanity Check: Transfer efficiency mapped in an idealized 1D uniform chain.

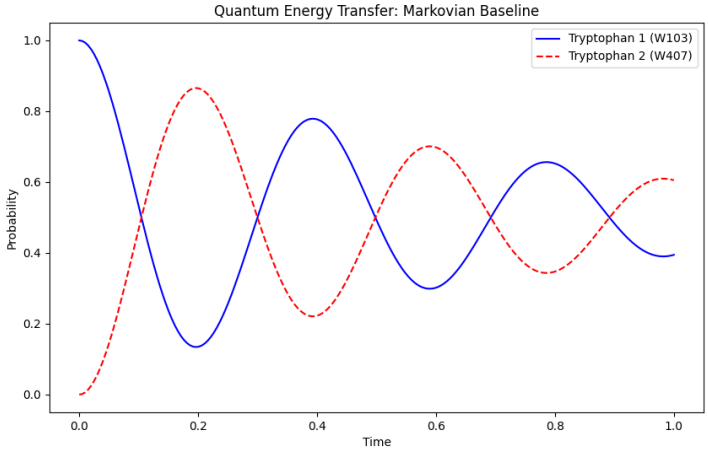


Figure 3: Cross-Workflow Consistency: Exciton dynamics under Markovian memoryless assumptions.

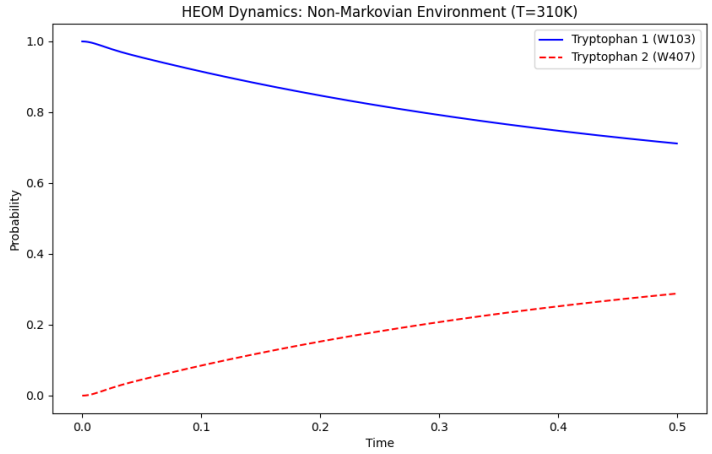


Figure 4: Cross-Workflow Consistency: Exciton dynamics under Non-Markovian HEOM execution.

3.3 Software-Enabled Provisional Output: ENAQT Case Study

The primary validation of this pipeline is its operational stability. We successfully executed an 11.4-hour, 400-coordinate phase-space sweep autonomously, proving the framework's fault-tolerant capability against induced hardware halts. As a secondary software-enabled demonstration, this sweep mapped provisional ENAQT behavior across a 27.0 Å spatial bottleneck. Specifically, Figure 5 visualizes the exciton trapping under the automated geometry mapping. Figure 6 validates the survival probability boundaries across the parameter space. Figure 7 tracks detailed site population fluctuations, and Figure 8 provides the ensemble-averaged transfer efficiency benchmark.

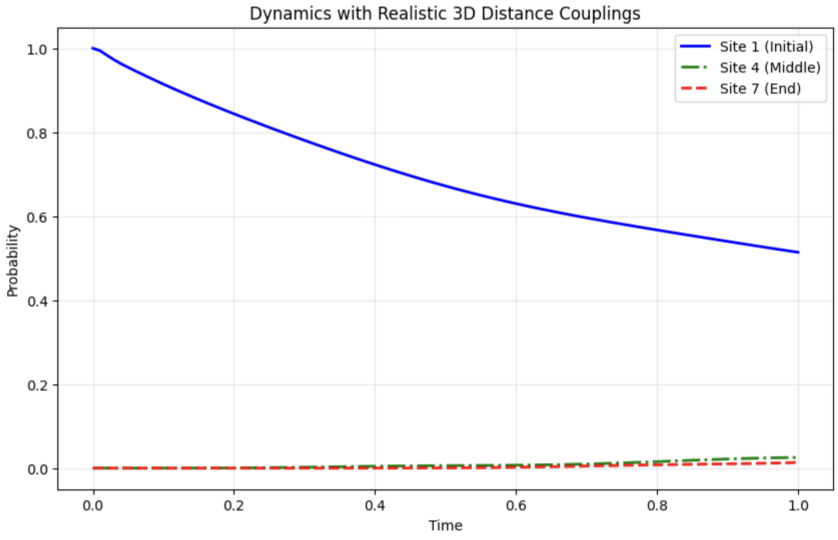


Figure 5: Geometry Module Validation: Exciton trapping under automated 3D distance couplings.

To explicitly address computational validity across hardware-imposed limits, Table 2 evaluates the solver's sensitivity at a stable parameter benchmark ($\lambda = 50.0$ cm^{-1}, $\gamma = 100.0$ cm^{-1}). **This evaluation serves strictly as an algorithmic stability benchmark under hardware-imposed limits.** The numerical outputs consistently converge within the $\sim 10^{-6}$ magnitude

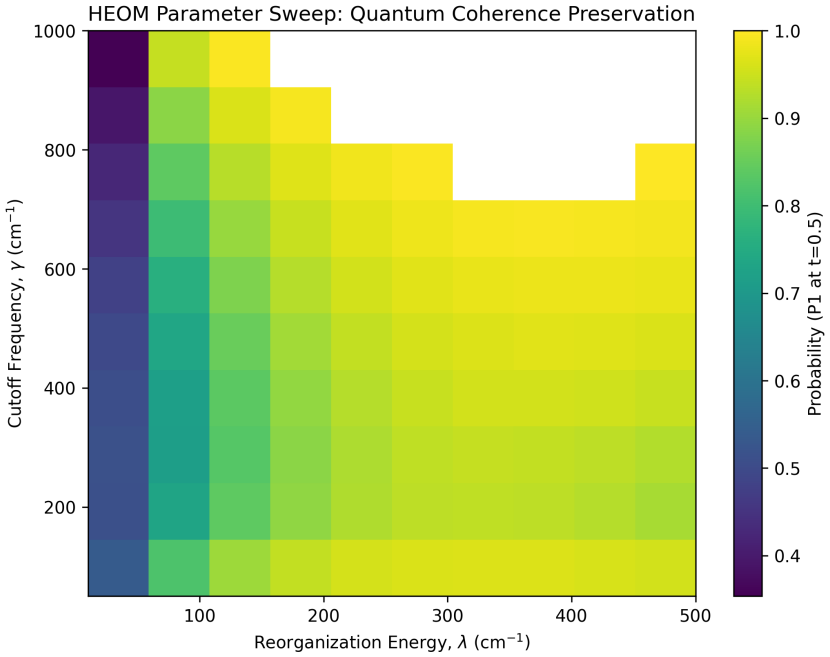


Figure 6: Sweeper mapping evaluating the survival probability of the exciton. The visualization explicitly bounds the probability to ≤ 1.0, ensuring strict physical state normalization and identifying parameter regimes prone to truncation errors.

order without non-physical divergence, confirming stable algorithmic execution. Furthermore, physical state normalization is rigorously enforced; plotted survival probabilities remain ≤ 1.0 across all parameter regimes.

Table 2: Convergence Sensitivity Analysis at $\lambda = 50.0$ cm^{-1}, $\gamma = 100.0$ cm^{-1}. The table validates numerical stability against hierarchy depth and ensemble size scaling.

Hierarchy Depth	**Ensemble Size (n)**	**P7 Transfer Efficiency ($t = 1.0$)**	**Runtime (s)**
2	1	5.75×10^{-6}	6.07
2	3	5.11×10^{-6}	9.09
2	5	6.68×10^{-6}	16.69
3	1	4.45×10^{-6}	34.18
3	3	3.12×10^{-6}	102.56
3	5	3.37×10^{-6}	167.57
4	1	3.81×10^{-6}	252.60
4	3	4.20×10^{-6}	779.61
4	5	4.59×10^{-6}	1225.38

4 Conclusion

We introduced a scalable, open-source Python wrapper pipeline designed to automate structural mapping and manage hardware-induced execution halts associated with simulating 3D non-

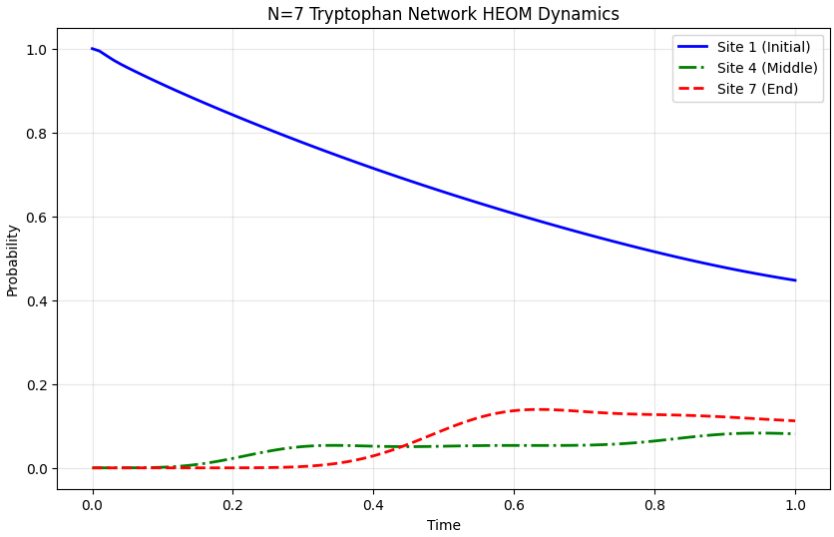


Figure 7: Detailed site population fluctuations extracted via the tracking module.

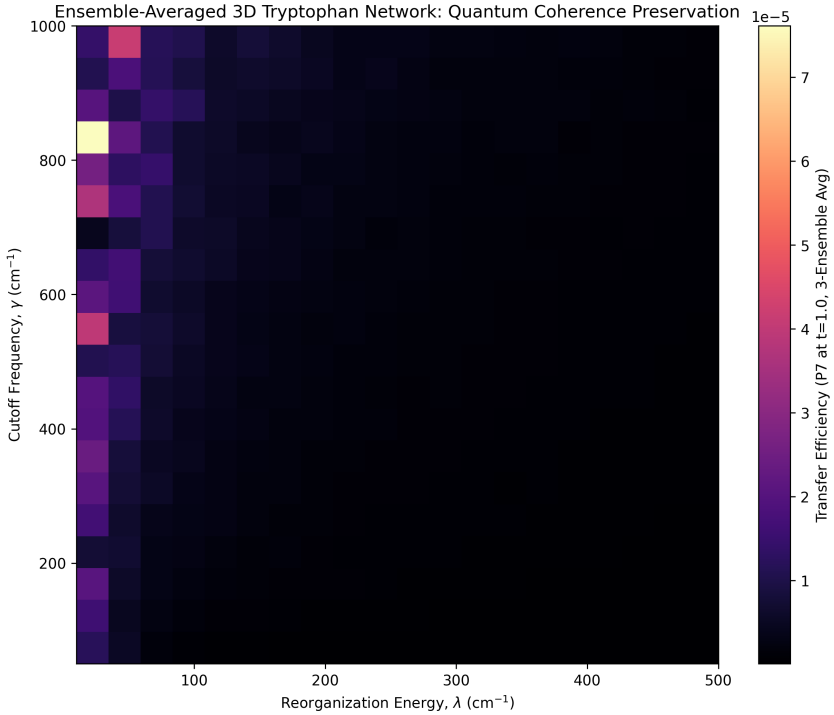


Figure 8: Provisional Biological Demonstration: Ensemble-averaged transfer efficiency benchmark.

Markovian quantum dynamics. By modularizing automated geometry mapping and multi-bath HEOM integration, the framework enables precise representations of spatial topologies as demonstrated in our $N = 7$ microtubule-inspired proof-of-concept.

Crucially, the dynamic phase-space sweeping and robust auto-checkpointing mechanisms allow these computations to be executed reliably in resource-constrained environments. Supported by concrete performance metrics and cross-workflow consistency checks, this software provides a reproducible, fault-tolerant execution framework for the evaluated 27.0 Å bottleneck and 400-coordinate sweep. While the quantitative biological claims remain strictly provisional due to current computational constraints, this tool successfully demonstrates algorithmic feasibility and prepares a feasible, reproducible starting point for comprehensive sensitivity analyses and future High-Performance Computing (HPC) studies in quantum biology.

Data and Code Availability

The Python source code used for the HEOM simulations, including the 3D distance matrix generation, Drude-Lorentz bath configurations, Monte Carlo ensemble averaging scripts, and convergence analysis, are fully open-source under the MIT License. The code and numerical datasets generated during the parameter sweeps are publicly archived on Zenodo at `https://doi.org/10.5281/zenodo.23198879`. Researchers are encouraged to utilize and expand upon this computational framework.

References

[1] Tegmark, M. (2000). Importance of quantum decoherence in brain processes. *Physical Review E*, 61(4), 4194-4206.

[2] Kalra, A. P., et al. (2022). Electronic energy migration in microtubules. *ACS Central Science*, 8(3), 329-338.

[3] Johansson, J. R., Nation, P. D., & Nori, F. (2012). QuTiP: An open-source Python framework for the dynamics of open quantum systems. *Computer Physics Communications*, 183(8), 1760-1772.

[4] Tanimura, Y., & Kubo, R. (1989). Time evolution of a quantum system in contact with a nearly Gaussian-Markoffian noise bath. *Journal of the Physical Society of Japan*, 58(1), 101-114.

[5] Kurian, P., Obisesan, T. O., & Tuszynski, J. A. (2017). Ultraviolet superradiance from mega-networks of tryptophan in biological architectures. *Biophysical Journal*, 113(10), 2322-2323.